

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 9: Singular Value Decomposition (SVD)

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1 Keywords

- Singular Valued Decomposition, SVD, general linear maps

2 Main Points

2.1 Singular Value Decomposition

- Any $n \times m$ matrix, $\mathbf{X}$ can be decomposed as $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T$
 - $\mathbf{U}$ is an $n \times n$ orthogonal matrix
 - $\mathbf{S}$ is an $n \times m$ matrix with zeros everywhere except the diagonal where $S_{ii} = s_i \geq 0$
 - $\mathbf{V}$ is an $m \times m$ orthogonal matrix
- The values s_i are known as the *singular values* of $\mathbf{X}$
- The SVD of a symmetric matrix is just the eigen-decomposition
- **Economical SVD**
 - If $n > m$ some algorithms won't bother outputting the last $n - m$ columns of $\mathbf{U}$
 - If $m < n$ some algorithms won't bother outputting the last $m - n$ columns of $\mathbf{V}$
 - In this case it will output a square $\mathbf{S}$ matrix

2.2 General Linear Mapping

- Recall that matrices are the most general linear operators
- Since any matrix $\mathbf{M}$ can be written as $\mathbf{U} \mathbf{S} \mathbf{V}^T$ we can interpret any linear mapping as doing three operations
 - A rotation (with possibly a reflection) defined by $\mathbf{V}^T$
 - A rescaling of each coordinate by s_i
 - A rotation (with possibly a reflection) defined by $\mathbf{U}$
- **Duality**
 - Using $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ then
 - $\mathbf{C} = \mathbf{X} \mathbf{X}^T = \mathbf{U} \mathbf{S} \mathbf{S}^T \mathbf{U}$
 - $\mathbf{D} = \mathbf{X}^T \mathbf{X} = \mathbf{V} \mathbf{S}^T \mathbf{S} \mathbf{V}$
 - $\mathbf{S} \mathbf{S}^T$ and $\mathbf{S}^T \mathbf{S}$ are diagonal elements with non-zero diagonal elements s_i^2

2.3 Ridge Regression

- Ridge regression is linear regression with an L_2 regulariser
- Adding a regulariser $\nu \|w\|^2$ the weights, w^* , that minimise the loss function are given by $w^* = (\mathbf{X}^T \mathbf{X} + \nu \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$
- Using $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ then

$$w^* = \mathbf{V} \bar{\mathbf{S}}^+ \mathbf{U}^T \mathbf{y}$$

where $\bar{\mathbf{S}}^+$ is a regularised pseudo-inverse of $\mathbf{S}$ given by

$$\bar{\mathbf{S}}^+ = (\mathbf{S}^T \mathbf{S} + \nu \mathbf{I})^{-1} \mathbf{S}$$

- If $\nu = 0$ this is equal to the pseudo-inverse of $\mathbf{S}$
- $\bar{\mathbf{S}}^+$ is an $n \times m$ matrix which is zero everywhere except on the diagonal, where $\bar{S}_{ii}^+ = \frac{s_i}{s_i^2 + \nu}$
 - Note if $s_i = 0$ linear regression has an infinity of solutions and the pseudo-inverse of $\mathbf{X}$ does not exist (setting $\nu = 0$ we get $S_{ii}^+ = 1/s_i$ which is not defined when $s_i = 0$)
 - In the regularised case $\bar{S}_{ii}^+ = 0$ (we have selected one of the solutions that minimise the squared error)
 - If $s_i \ll \nu$ then without the regularisation term the inverse is very ill-conditioned while with the regularisation term $\bar{S}_{ii}^+$ will be small
 - If $s_i \gg \nu$ then $\bar{S}_{ii}^+ \approx \frac{1}{s_i} = S_{ii}^+$
- Adding a L_2 regulariser means that the optimum weights, w^* , will be less sensitive to the training data reducing the variance in the bias-variance dilemma

3 Exercises

3.1 Ridge regression

- Ridge regression is just linear regression with an L_2 regulariser
 1. Derive the optimal weights in ridge regression
 2. Show that using $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ then $w^* = \mathbf{V} (\mathbf{S}^T \mathbf{S} + \nu \mathbf{I})^{-1} \mathbf{S} \mathbf{U} \mathbf{y}$
- See answers

4 Experiments

4.1 SVD

- Using Matlab/Octave or python have a play with svd

```
X = randn(3,4)           % construct a random matrix
[U,S,V] = svd(X)         % compute singular value decomposition
U*S*V'                   % should be the same as X
U*U'                     % should be the identity up to round error
U'*U                     % should be the identity up to round error
V*V'                     % should be the identity up to round error
V'*V                     % should be the identity up to round error
[Ue,L1] = eig(X*X')      % Ue should be the same as U up to permutation
S*S'                     % same as L1 up to permutation
```

```
[Ve,L2] = eig(X'*X) % Ve should be the same as V up to permutation
S'*S          % same as L2 up to permutation
```

```
inv(X'*X + 0.1*eye(4)) % check identity
V*inv(S'*S + 0.1*eye(4))*V' % should be the same
```

4.2 Verify Identity

- Again use Matlab/Octave or python
- For a random 4×5 matrix $\mathbf{X}$
 - Check that using $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ that

$$(\mathbf{X}^T\mathbf{X} + \eta\mathbf{I})^{-1} = \mathbf{V}(\mathbf{S}^T\mathbf{S} + \nu\mathbf{I})^{-1}\mathbf{V}^T$$

holds for some random matrix using Matlab/Octave or python

- Examine $\mathbf{S}^T\mathbf{S}$, $\mathbf{S}^T\mathbf{S} + 0.1\mathbf{I}$, $(\mathbf{S}^T\mathbf{S} + 0.1\mathbf{I})^{-1}$ and $(\mathbf{S}^T\mathbf{S} + 0.1\mathbf{I})^{-1}\mathbf{S}^T$
- See if you can invert $\mathbf{X}^T\mathbf{X}$: it is singular, but due to rounding errors it might be inverted (it was a scary matrix when I tried it)

```
X = randn(4,5) % construct a random matrix
[U,S,V] = svd(X) % compute singular value decomposition
```

```
inv(X'*X + 0.1*eye(5)) % check identity
V*inv(S'*S + 0.1*eye(5))*V' % should be the same
```

```
S'*S % singular
S'*S + 0.1*eye(5) % now invertible
inv(S'*S + 0.1*eye(5))
inv(S'*S + 0.1*eye(5))*S' % 4x5 diagonal matrix
```

```
inv(X'*X) % shouldn't be able to do this
```

5 Answers

5.1 Ridge regression

1. It is straightforward to show

$$\mathbf{w}^* = (\mathbf{X}^T\mathbf{X} + \nu\mathbf{I})^{-1}\mathbf{X}^{-1}\mathbf{y}$$

2. The only hard part is to show is that

$$(\mathbf{X}^T\mathbf{X} + \nu\mathbf{I})^{-1} = \mathbf{V}(\mathbf{S}^T\mathbf{S} + \nu\mathbf{I})^{-1}\mathbf{V}^T$$

- It is easy to show that $\mathbf{X}^T\mathbf{X} = \mathbf{V}\mathbf{S}^T\mathbf{S}\mathbf{V}^T$
- But we also have $\mathbf{I} = \mathbf{V}\mathbf{V}^T$ as $\mathbf{V}$ is an orthogonal matrix
- Thus $\mathbf{M} = \mathbf{X}^T\mathbf{X} + \nu\mathbf{I} = \mathbf{V}(\mathbf{S}^T\mathbf{S} + \nu\mathbf{I})\mathbf{V}^T = \mathbf{V}\mathbf{W}\mathbf{V}^T$ where $\mathbf{W} = \mathbf{S}^T\mathbf{S} + \nu\mathbf{I}$
- But $(\mathbf{A}\mathbf{B}\mathbf{C})^{-1} = \mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1}$ (which we can verify by multiplying $\mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1}$ on either the left or right by $\mathbf{A}\mathbf{B}\mathbf{C}$)
- Thus $\mathbf{M}^{-1} = (\mathbf{V}\mathbf{W}\mathbf{V}^T)^{-1} = (\mathbf{V}^T)^{-1}\mathbf{W}^{-1}\mathbf{V}^{-1} = \mathbf{V}\mathbf{W}^{-1}\mathbf{V}^T$ where we use $\mathbf{V}^{-1} = \mathbf{V}^T$ as $\mathbf{V}$ is an orthogonal matrix